

L3: Introduction to Linear Regression

Topics to be covered today:

- Supervised learning setting
- Linear regression formulation
- Closed form solution
- Geometric interpretation
- Case study

Supervised Learning Setting

Recall: We are given n training samples:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Training data

Here, $x_i \in \mathbb{R}^p$, i.e., each input is a p -dimensional vector.

★ The components of x_i are also called features.
We assume for now that the features are quantitative.

Assume for now that $y_i \in \mathbb{R}$, i.e., the o/p _{$\wedge$} is scalar. Variable

Goal: Learn, from the training data, a "good" input-output relationship, i.e., learn f s.t.

$$\hat{y} = f(x)$$

is a "good" prediction of the o/p for an unseen input x .
{ In practice, we look for f that "fits" the training data well. }

Example:

Input = (TV Ad spend, Radio Ad spend, Newspapers Ad spend)

(3 features; $p = 3$)

Output = Sales

ADVERTISING
DATASET
from
ISL

We typically vectorize the training data as follows:

$$X = \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{pmatrix}_{n \times p}$$

Rows of X : training inputs

Columns of X : features

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1}$$

The training data is described by the $n \times p$ matrix X & the $n \times 1$ column vector Y .

training input

Typically, $n \gg p$, making X tall.

training output

Linear Regression

In this approach, we try to learn/fit a linear input-output relationship.

That is,

$$f(x) = \hat{\beta}_0 + \sum_{i=1}^p \hat{\beta}_i x(i)$$

$\hat{\beta}_0$ is labeled as *intercept* with a green arrow pointing to it.

The vector $\begin{pmatrix} x(1) \\ x(2) \\ \vdots \\ x(p) \end{pmatrix}$ is labeled with a green arrow pointing to it.

The coefficients $\hat{\beta}_i$ are labeled with a green arrow pointing to them as *coefficient corresponding to i th feature*.

Note: $\hat{\beta}_0$ is the predicted o/p for "zero" input

$\hat{\beta}_i$ the predicted change in o/p from a unit increase in feature i

We can rewrite the above linear relationship as follows:

$$f(x) = \left\langle \begin{pmatrix} 1 \\ x(1) \\ \vdots \\ x(p) \end{pmatrix}, \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{pmatrix} \right\rangle$$

The first vector is labeled with a green wavy line as *input with constant feature appended at the top*.

The second vector is labeled with a green wavy line as *regression parameters to be fit*.

Q : How do we fit the regression coefficients?

A : Try to minimize error over the training data.

Specifically, we minimize

$$\text{RSS}(\beta) = \sum_{i=1}^n \left(y_i - \left\langle \begin{pmatrix} 1 \\ x_i(1) \\ \vdots \\ x_i(p) \end{pmatrix}, \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \right\rangle \right)^2$$

Residual sum of squares

Squared error between y_i & predicted o/p corresponding to x_i

This quantity can also be expressed in vector notation:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_1^T \\ 1 & x_2^T \\ \vdots & \vdots \\ 1 & x_n^T \end{pmatrix}_{n \times (p+1)}$$

$$\Rightarrow \text{RSS}(\beta) = \langle Y - X\beta, Y - X\beta \rangle$$

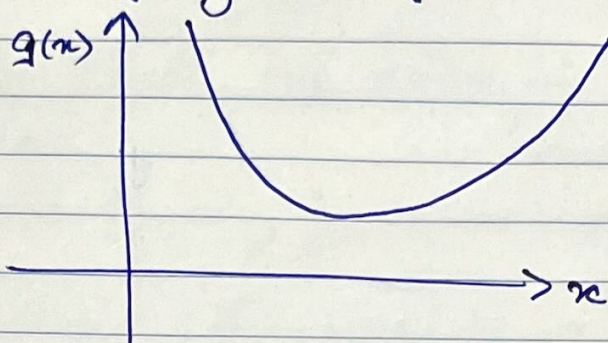
$$= \|Y - X\beta\|^2$$

vector of predicted outputs

Note: $RSS(\beta)$ is a quadratic function of the regression coefficients.

Fact: $RSS(\cdot)$ is a convex function.

Convex functions are easy to minimize.
Minimizer (global) can be obtained by setting gradient (derivative) to zero.



A convex function
in 1-d

Minimizer of RSS has a clean closed form when X has full column rank.

↳ Columns of X are linearly independent

$$\text{rank}(X) = p+1$$

⇒ X is not wide
(typically X is very tall)

Formally $\hat{\beta}$ solves: $\min_{\beta \in \mathbb{R}^{p+1}} RSS(\beta)$

If X has full col. rank,

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Note: This that this is dimensionally legal.

$$X \sim n \times (p+1)$$

$$\Rightarrow X^T X \sim (p+1) \times (p+1)$$

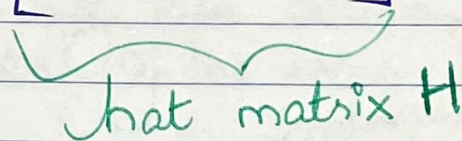
$$\Rightarrow (X^T X)^{-1} X^T \sim (p+1) \times n$$

$$\Rightarrow (X^T X)^{-1} X^T Y \sim (p+1) \times 1$$

In other words, the vector of predicted outputs is:

$$\hat{Y} = X \hat{\beta}$$

$$= \left[X (X^T X)^{-1} X^T \right] Y$$

hat matrix H

Geometric interpretation:

We have sought to minimize

$$\|Y - \hat{Y}\|^2, \text{ where}$$

$$\hat{y} = X\beta = \begin{bmatrix} \mathbb{1} & X_1 & X_2 & \cdots & X_p \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

columns of X
 $X_i \sim i^{\text{th}}$ feature
 across training data

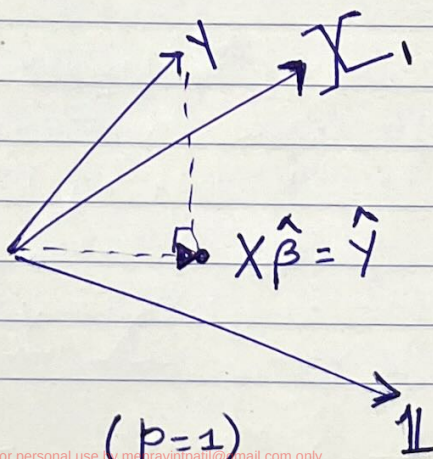
$$= \mathbb{1}\beta_0 + X_1\beta_1 + \cdots + X_p\beta_p$$

Moral: • $\hat{y}$ is a linear combination of the columns of X

• In effect, we are finding the best linear combination of these columns, that minimizes norm square of the error $\|y - \hat{y}\|^2$.

• $\hat{\beta}$ is simply the ^(vectorized) collection of coefficients of this optimal linear combination.

$\hat{y}$ is the projection of y on to the column space of X .



Takeaways:

- $\hat{Y} = X\hat{\beta} = HY$ is the projection of Y on to the column space of X .
- The hat matrix H performs this ^{very} projection!